

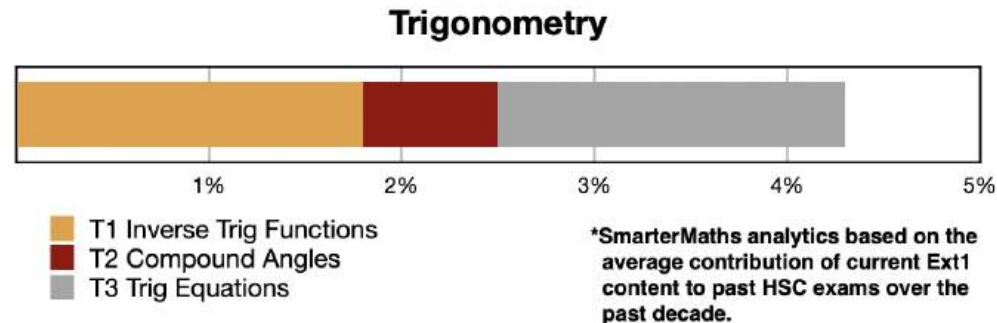
EXT 1: Trigonometry (Ext1), T3 Trig Equations (Y12)

Auxiliary Angles (Ext1)

Identities, Equations and 't' formulae (Ext1)

Teacher: Ljiljana Todoc

Exam Equivalent Time: 127.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)



HISTORICAL CONTRIBUTION

- *Trigonometry* looks at trigonometric algebra, graphs and identities. This is an important area that is examined both in its own right and used within calculus applications that are covered in later topics.
- This topic has been split into three sub-categories for analysis purposes which are: 1-Inverse Trig Functions (1.8%), 2-Compound Angles (0.7%) and 3-Trig Equations (1.8%).
- This analysis looks at *Trig Equations - Proving Identities and 't' formulae*.

HSC ANALYSIS - What to expect and common pitfalls

- *Proving Identities and 't' formulae* is a smaller sub-topic than *Auxiliary Angles* within *T3 Trig Equations*.
- *Proving Identities* was examined in a big way in 2020. It was allocated a significant 4-marks within a very challenging 7-mark cross topic question, and should be carefully reviewed (see 2020 Ext1 14b).
- Prior to 2020, it was given a dedicated question each year between 2008-10, with 2008 Ext1 6b causing students real problems.
- A revision focus here is important. The use of compound and double angle identities is not only core knowledge for this topic area, but it will also prove *extremely* valuable in later applications of calculus.
- NESA Topic Guidance explicitly states that students can be asked to provide "all" solutions to some trig equations. Carefully review *T3 EQ-Bank 3* to see suggested answers that do not require knowledge of the general solution formulae.
- The 't' formulae were last examined in 2007 and 2005. Its inclusion in the new syllabus make it a smoky bear in our view - ensure you revise and understand this area!

Questions

1. Trigonometry, EXT1 T3 2014 HSC 2 MC

Which expression is equal to $\cos x - \sin x$?

- (A) $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right)$
- (B) $\sqrt{2} \cos\left(x - \frac{\pi}{4}\right)$
- (C) $2 \cos\left(x + \frac{\pi}{4}\right)$
- (D) $2 \cos\left(x - \frac{\pi}{4}\right)$

2. Trigonometry, EXT1 T3 2017 HSC 4 MC

What is the value of $\tan \alpha$ when the expression $2 \sin x - \cos x$ is written in the form $\sqrt{5} \sin(x - \alpha)$?

- A. -2
- B. $-\frac{1}{2}$
- C. $\frac{1}{2}$
- D. 2

3. Trigonometry, EXT1 T3 2015 HSC 11d

Express $5 \cos x - 12 \sin x$ in the form $A \cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$. (2 marks)

4. Trigonometry, EXT1 T3 2018 HSC 11c

Write $\sqrt{3} \sin x + \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. (2 marks)

5. Trigonometry, EXT1 T3 SM-Bank 10

Given that $\cot(2x) + \frac{1}{2} \tan(x) = a \cot(x)$, calculate a . (3 marks)

6. Calculus, EXT1 C2 2005 HSC 3b

i. By expanding the left-hand side, show that

$$\sin(5x + 4x) + \sin(5x - 4x) = 2 \sin 5x \cos 4x \quad (1 \text{ mark})$$

ii. Hence find $\int \sin 5x \cos 4x \, dx$. *(2 marks)*

7. Trigonometry, EXT1 T3 2013 HSC 12a

i. Write $\sqrt{3} \cos x - \sin x$ in the form $2 \cos(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. *(1 mark)*

ii. Hence, or otherwise, solve $\sqrt{3} \cos x = 1 + \sin x$, where $0 < x < 2\pi$. *(2 marks)*

8. Trigonometry, EXT1 T3 2004 HSC 2d

i. Write $8 \cos x + 6 \sin x$ in the form $A \cos(x - \alpha)$, where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. *(2 marks)*

ii. Hence, or otherwise, solve the equation $8 \cos x + 6 \sin x = 5$ for $0 \leq x \leq 2\pi$.
Give your answers correct to three decimal places. *(2 marks)*

9. Trigonometry, EXT1 T3 2007 HSC 2a

By using the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$. *(2 marks)*

10. Trigonometry, EXT1 T3 2009 HSC 2b

i. Express $3 \sin x + 4 \cos x$ in the form $A \sin(x + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$. *(2 marks)*

ii. Hence, or otherwise, solve $3 \sin x + 4 \cos x = 5$ for $0 \leq x \leq 2\pi$.
Give your answer, or answers, correct to two decimal places. *(2 marks)*

11. Trigonometry, EXT1 T3 2009 HSC 3c

i. Prove that $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$ provided that $\cos 2\theta \neq -1$. *(2 marks)*

ii. Hence find the exact value of $\tan \frac{\pi}{8}$. *(1 mark)*

12. Trigonometry, EXT1 T3 2010 HSC 6a

i. Show that $\cos(A - B) = \cos A \cos B(1 + \tan A \tan B)$. *(1 mark)*

ii. Suppose that $0 < B < \frac{\pi}{2}$ and $B < A < \pi$.

Deduce that if $\tan A \tan B = -1$, then $A - B = \frac{\pi}{2}$. *(1 mark)*

13. Trigonometry, EXT1 T3 SM-Bank 13

Find all solutions of $\tan(2\theta) = -\tan \theta$ for $0 \leq \theta \leq 2\pi$. *(3 marks)*

14. Trigonometry, EXT1 T3 SM-Bank 11

Given that $\cos(\theta - \phi) = \frac{3}{5}$ and $\tan \theta \tan \phi = 2$, find $\cos(\theta + \phi)$. *(3 marks)*

15. Trigonometry, EXT1 T3 SM-Bank 12

Solve $\sin(2x) = \sin x$, for $0 \leq x \leq 2\pi$. *(3 marks)*

16. Trigonometry, EXT1 T3 SM-Bank 8

Solve for x , given that

$$x \sin x \sec(2x) = 0, \quad 0 \leq x \leq 2\pi \quad (2 \text{ marks})$$

17. Trigonometry, EXT1 T3 2020 HSC 11d

By expressing $\sqrt{3} \sin x + 3 \cos x$ in the form $A \sin(x + a)$, solve $\sqrt{3} \sin x + 3 \cos x = \sqrt{3}$, for $0 \leq x \leq 2\pi$. *(4 marks)*

18. Trigonometry, EXT1 T3 2021 HSC 11g

By factorising, or otherwise, solve $2 \sin^3 x + 2 \sin^2 x - \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$. *(3 marks)*

19. Trigonometry, EXT1 T3 EQ-Bank 3

Find all values of θ that satisfy the equation $\sqrt{3} \cos \theta = \sin(2\theta)$. *(3 marks)*

20. Trigonometry, EXT1 T3 EQ-Bank 4

The current flowing through an electrical circuit can be modelled by the function

$$f(t) = 6 \sin 0.05t + 8 \cos 0.05t, \quad t \geq 0$$

- Express the function in the form $f(t) = A \sin(at + b)$, for $0 \leq b \leq \frac{\pi}{2}$. (2 marks)
- Find the time at which the current first obtains its maximum value. (1 mark)
- Sketch the graph of $f(t)$. Clearly show its range and label the coordinates of its first maximum value. Do not label x -intercepts. (1 mark)

21. Trigonometry, EXT1 T3 SM-Bank 2

Show that

$$\cos 3x = 4 \cos^3 x - 3 \cos x. \quad (3 \text{ marks})$$

22. Trigonometry, EXT1 T3 2010 HSC 4b

- Express $2 \cos \theta + 2 \cos\left(\theta + \frac{\pi}{3}\right)$ in the form $R \cos(\theta + \alpha)$,
where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. (3 marks)
- Hence, or otherwise, solve $2 \cos \theta + 2 \cos\left(\theta + \frac{\pi}{3}\right) = 3$,
for $0 < \theta < 2\pi$. (2 marks)

23. Trigonometry, EXT1 T3 EQ-Bank 1

- Show that $\sin x + \sin 3x = 2 \sin 2x \cos x$. (2 marks)
- Hence or otherwise, find all values of x that satisfy

$$\sin x + \sin 2x + \sin 3x = 0, \quad x \in [0, 2\pi]. \quad (2 \text{ marks})$$

24. Trigonometry, EXT1 T3 EQ-Bank 2

A particular energy wave can be modelled by the function

$$f(t) = \sqrt{5} \sin 0.2t + 2 \cos 0.2t, \quad t \in [0, 50]$$

- Express this function in the form $f(t) = R \sin(nt - \alpha)$, $\alpha \in [0, 2\pi]$. (2 marks)
- Find the time the wave first attains its maximum value. Give your answer to one decimal place. (2 marks)

25. Trigonometry, EXT1 T3 2005 HSC 4b

By making the substitution $t = \tan \frac{\theta}{2}$, or otherwise, show that

$$\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}. \quad (2 \text{ marks})$$

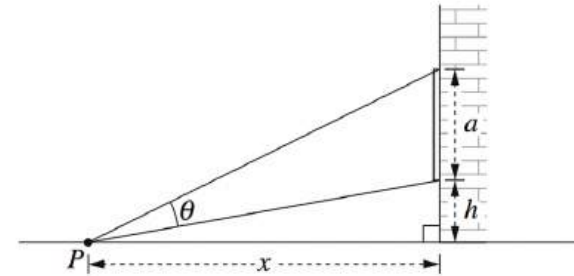
26. Trigonometry, EXT1 T3 2008 HSC 6b

It can be shown that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ for all values of θ . (Do NOT prove this.)

Use this result to solve $\sin 3\theta + \sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$. (3 marks)

27. Trigonometry, EXT1 T3 SM-Bank 1

A billboard of height a metres is mounted on the side of a building, with its bottom edge h metres above street level. The billboard subtends an angle θ at the point P , x metres from the building.



Use the identity $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$ to show that

$$\theta = \tan^{-1} \left[\frac{ax}{x^2 + h(a + h)} \right]. \quad (2 \text{ marks})$$

28. Trigonometry, EXT1 T3 2020 HSC 14b

- Show that $\sin^3 \theta - \frac{3}{4} \sin \theta + \frac{\sin(3\theta)}{4} = 0$. (2 marks)
- By letting $x = 4 \sin \theta$ in the cubic equation $x^3 - 12x + 8 = 0$.
Show that $\sin(3\theta) = \frac{1}{2}$. (2 marks)
- Prove that $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{3}{2}$. (3 marks)

Worked Solutions

1. Trigonometry, EXT1 T3 2014 HSC 2 MC

$$R \cos(x + \alpha) = R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$\therefore R \cos x \cos \alpha - R \sin x \sin \alpha = \cos x - \sin x$$

$$R \cos \alpha = 1, \quad R \sin \alpha = 1$$

$$R^2 = 1^2 + 1^2$$

$$R = \sqrt{2}$$

$$\therefore \sqrt{2} \cos \alpha = 1$$

$$\cos \alpha = \frac{1}{\sqrt{2}}$$

$$\alpha = \frac{\pi}{4}$$

$$\therefore \sqrt{2} \cos\left(x + \frac{\pi}{4}\right) = \cos x - \sin x$$

$$\Rightarrow A$$

2. Trigonometry, EXT1 T3 2017 HSC 4 MC

$$\text{Using } \sqrt{5} \sin(x - \alpha) = \sqrt{5} \sin x \cos \alpha - \sqrt{5} \cos x \sin \alpha,$$

$$\Rightarrow \sqrt{5} \sin x \cos \alpha - \sqrt{5} \cos x \sin \alpha = 2 \sin x - \cos x$$

$$\Rightarrow \sqrt{5} \cos \alpha = 2, \sqrt{5} \sin \alpha = 1$$

$$\frac{\sin \alpha}{\cos \alpha} = \frac{1}{2}$$

$$\therefore \tan \alpha = \frac{1}{2}$$

$$\Rightarrow C$$

3. Trigonometry, EXT1 T3 2015 HSC 11d

$$5 \cos x - 12 \sin x = A \cos(x + \alpha)$$

$$= A \cos x \cos \alpha - A \sin x \sin \alpha$$

$$\therefore A \cos \alpha = 5, \quad A \sin \alpha = 12$$

$$A^2 = 5^2 + 12^2 = 169$$

$$A = 13$$

$$\Rightarrow 13 \cos \alpha = 5$$

$$\cos \alpha = \frac{5}{13}$$

$$\alpha = \cos^{-1} \left(\frac{5}{13} \right) \approx 1.176 \dots \text{radians}$$

$$\therefore 5 \cos x - 12 \sin x = 13 \cos(x + 1.176 \dots)$$

4. Trigonometry, EXT1 T3 2018 HSC 11c

$$R \sin(x + \alpha) = \sqrt{3} \sin x + \cos x$$

$$R \sin x \cos \alpha + R \cos x \sin \alpha = \sqrt{3} \sin x + \cos x$$

Equating coefficients:

$$\Rightarrow R \cos \alpha = \sqrt{3}$$

$$\Rightarrow R \sin \alpha = 1$$

$$R^2 = (\sqrt{3})^2 + 1^2 = 4$$

$$\therefore R = 2$$

$$\Rightarrow 2 \sin \alpha = 1$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{6} \quad \left(0 \leq \alpha \leq \frac{\pi}{2} \right)$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin \left(x + \frac{\pi}{6} \right).$$

5. Trigonometry, EXT1 T3 SM-Bank 10

$$\frac{1}{\tan(2x)} + \frac{\tan(x)}{2} = \frac{a}{\tan(x)}, \quad \tan(x) \neq 0$$

$$\frac{1 - \tan^2(x)}{2 \tan(x)} + \frac{\tan(x)}{2} = \frac{a}{\tan(x)}$$

If $\tan(x) \neq 0$:

$$\frac{1 - \tan^2(x) + \tan^2(x)}{2 \tan(x)} = \frac{a}{\tan(x)}$$

$$1 = 2a$$

$$\therefore a = \frac{1}{2}$$

6. Calculus, EXT1 C2 2005 HSC 3b

i. $\sin(5x + 4x) + \sin(5x - 4x) = 2 \sin 5x \cos 4x$

$$\begin{aligned} \text{LHS} &= \sin 5x \cos 4x - \sin 4x \cos 5x + \sin 5x \cos 4x \\ &\quad + \sin 4x \cos 5x \\ &= 2 \sin 5x \cos 4x \dots \text{as required} \end{aligned}$$

ii. $\int \sin 5x \cos 4x \, dx$

$$= \frac{1}{2} \int 2 \sin 5x \cos 4x \, dx$$

$$= \frac{1}{2} \int \sin(5x + 4x) + \sin(5x - 4x) \, dx$$

$$= \frac{1}{2} \int \sin 9x + \sin x \, dx$$

$$= \frac{1}{2} \left[-\frac{1}{9} \cos 9x - \cos x \right] + c$$

$$= -\frac{1}{18} \cos 9x - \frac{1}{2} \cos x + c$$

7. Trigonometry, EXT1 T3 2013 HSC 12a

i. Write $\sqrt{3} \cos x - \sin x$ in form

$$2 \cos(x + \alpha), \quad 0 < \alpha < \frac{\pi}{2}$$

$$2(\cos x \cos \alpha - \sin x \sin \alpha) = \sqrt{3} \cos x - \sin x$$

$$\cos x \cos \alpha - \sin x \sin \alpha = \frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x$$

$$\Rightarrow \cos \alpha = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sin \alpha = \frac{1}{2}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore 2 \cos\left(x + \frac{\pi}{6}\right) = \sqrt{3} \cos x - \sin x$$

ii. Solve $\sqrt{3} \cos x = 1 + \sin x$, $0 < x < 2\pi$

$$\sqrt{3} \cos x - \sin x = 1$$

$$2 \cos\left(x + \frac{\pi}{6}\right) = 1 \quad (\text{from part (i)})$$

$$\cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Since cos is positive in 1st / 4th quadrants

$$x + \frac{\pi}{6} = \frac{\pi}{3}, 2\pi - \frac{\pi}{3} \quad (0 < x < 2\pi)$$

$$\therefore x = \frac{\pi}{6}, \frac{3\pi}{2}$$

8. Trigonometry, EXT1 T3 2004 HSC 2d

i. Write $8 \cos x + 6 \sin x$ in the form $A \cos(x - \alpha)$

$$A(\cos x \cos \alpha + \sin x \sin \alpha) = 8 \cos x + 6 \sin x$$

$$(\cos x \cos \alpha + \sin x \sin \alpha) = \frac{8}{A} \cos x + \frac{6}{A} \sin x$$

$$\Rightarrow \cos \alpha = \frac{8}{A}$$

$$\Rightarrow \sin \alpha = \frac{6}{A}$$

$$\left(\frac{8}{A}\right)^2 + \left(\frac{6}{A}\right)^2 = 1$$

$$8^2 + 6^2 = A^2$$

$$\therefore A = \sqrt{100}$$

$$= 10$$

$$\Rightarrow \cos \alpha = \frac{8}{10}$$

$$\therefore \alpha = 0.6435 \dots \text{radians}$$

$$\therefore 8 \cos x + 6 \sin x = 10 \cos(x - 0.6435 \dots)$$

ii. $8 \cos x + 6 \sin x = 5$

$$10 \cos(x - 0.6435 \dots) = 5$$

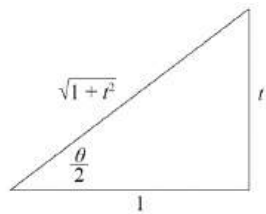
$$\cos(x - 0.6435 \dots) = \frac{1}{2}$$

$$x - 0.6435 \dots = \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$$

$$x = \frac{\pi}{3} + 0.6435 \dots \text{ or } \frac{5\pi}{3} + 0.6435 \dots$$

$$= 1.691 \text{ or } 5.879 \text{ radians (to 3 d.p.)}$$

9. Trigonometry, EXT1 T3 2007 HSC 2a



COMMENT: The values of $\sin \theta$ and $\cos \theta$ can be committed to memory or quickly derived using double angle formulae (as shown in the worked solution).

$$\Rightarrow \tan \frac{\theta}{2} = t$$

$$\Rightarrow \sin \theta = 2 \cdot \frac{t}{\sqrt{1+t^2}} \cdot \frac{1}{\sqrt{1+t^2}} = \frac{2t}{1+t^2}$$

$$\Rightarrow \cos \theta = \frac{1}{1+t^2} - \frac{t^2}{1+t^2} = \frac{1-t^2}{1+t^2}$$

Show $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$:

$$\begin{aligned} \frac{1 - \cos \theta}{\sin \theta} &= \frac{1 - \left(\frac{1-t^2}{1+t^2}\right)}{\frac{2t}{1+t^2}} \times \frac{1+t^2}{1+t^2} \\ &= \frac{1+t^2 - (1-t^2)}{2t} \\ &= \frac{2t^2}{2t} \\ &= t \\ &= \tan \frac{\theta}{2} \dots \text{as required} \end{aligned}$$

10. Trigonometry, EXT1 T3 2009 HSC 2b

i. $A \sin(x + \alpha) = 3 \sin x + 4 \cos x$

$$A \sin x \cos \alpha + A \cos x \sin \alpha = 3 \sin x + 4 \cos x$$

$$\Rightarrow A \cos \alpha = 3 \quad A \sin \alpha = 4$$

$$A^2 = 3^2 + 4^2 = 25$$

$$\therefore A = 5$$

$$\Rightarrow 5 \cos \alpha = 3$$

$$\cos \alpha = \frac{3}{5}$$

$$\alpha = \cos^{-1}\left(\frac{3}{5}\right) = 0.9272\dots \text{ radians}$$

$$\therefore 3 \sin x + 4 \cos x = 5 \sin\left(x + \cos^{-1}\left(\frac{3}{5}\right)\right)$$

ii. $3 \sin x + 4 \cos x = 5$

$$5 \sin(x + \alpha) = 5$$

$$\sin(x + \alpha) = 1$$

$$x + \alpha = \frac{\pi}{2}, \frac{5\pi}{2}, \dots$$

$$x = \frac{\pi}{2} - 0.9272\dots \quad (0 \leq x \leq 2\pi)$$

$$= 0.6435\dots$$

$$= 0.64 \text{ radians (2 d.p.)}$$

11. Trigonometry, EXT1 T3 2009 HSC 3c

i. Prove $\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$, $\cos 2\theta \neq -1$

$$\begin{aligned}\text{Using } \cos 2\theta &= 2\cos^2 \theta - 1 \\ &= 1 - 2\sin^2 \theta\end{aligned}$$

$$\begin{aligned}\text{RHS} &= \frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)} \\ &= \frac{2\sin^2 \theta}{2\cos^2 \theta} \\ &= \tan^2 \theta \dots \text{as required.}\end{aligned}$$

ii. $\tan^2 \frac{\pi}{8} = \frac{1 - \cos(2 \times \frac{\pi}{8})}{1 + \cos(2 \times \frac{\pi}{8})}$

$$\begin{aligned}&= \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1} \\ &= \frac{(\sqrt{2} - 1)^2}{2 - 1} \\ &= (\sqrt{2} - 1)^2\end{aligned}$$

$$\therefore \tan \frac{\pi}{8} = \sqrt{2} - 1 \quad \left(\tan\left(\frac{\pi}{8}\right) > 0 \right)$$

$$\left(\tan \frac{\pi}{8} = \sqrt{3 - 2\sqrt{2}} \text{ is also a correct answer} \right)$$

12. Trigonometry, EXT1 T3 2010 HSC 6a

i. Show $\cos(A - B) = \cos A \cos B(1 + \tan A \tan B)$

$$\begin{aligned}\text{RHS} &= \cos A \cos B \left(1 + \frac{\sin A \sin B}{\cos A \cos B} \right) \\ &= \cos A \cos B + \sin A \sin B \\ &= \cos(A - B) \dots \text{as required}\end{aligned}$$

ii. Given $\tan A \tan B = -1$

$$\cos(A - B) = \cos A \cos B(1 - 1)$$

$$\cos(A - B) = 0$$

$$\begin{aligned}A - B &= \cos^{-1} 0 \\ &= \frac{\pi}{2}, \frac{3\pi}{2}, \dots\end{aligned}$$

$$\text{Since } 0 < B < \frac{\pi}{2} \text{ and } B < A < \pi,$$

$$\Rightarrow A - B = \frac{\pi}{2}$$

13. Trigonometry, EXT1 T3 SM-Bank 13

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = -\tan \theta$$

Let $\tan \theta = k$,

$$\frac{2k}{1 - k^2} = -k$$

$$2k = -k(1 - k^2)$$

$$2k + k(1 - k^2) = 0$$

$$3k - k^3 = 0$$

$$k(3 - k^2) = 0$$

$$k(3 - k^2) = 0 \Rightarrow k = 0, k = \pm \sqrt{3}$$

If $\tan \theta = 0 \Rightarrow \theta = 0, \pi, 2\pi$

If $\tan \theta = \pm \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

$$\therefore \theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi$$

14. Trigonometry, EXT1 T3 SM-Bank 11

$$\cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$\cos \theta \cos \phi + \sin \theta \sin \phi = \frac{3}{5} \dots (1)$$

$$\frac{\sin \theta \sin \phi}{\cos \theta \cos \phi} = 2$$

$$\sin \theta \sin \phi = 2 \cos \theta \cos \phi \dots (2)$$

Substitute (2) into (1):

$$\cos \theta \cos \phi + 2 \cos \theta \cos \phi = \frac{3}{5}$$

$$3 \cos \theta \cos \phi = \frac{3}{5}$$

$$\cos \theta \cos \phi = \frac{1}{5}$$

Substitute $\cos \theta \cos \phi = \frac{1}{5}$ into (1):

$$\frac{1}{5} + \sin \theta \sin \phi = \frac{3}{5}$$

$$\sin \theta \sin \phi = \frac{2}{5}$$

$$\therefore \cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$= \frac{1}{5} - \frac{2}{5}$$

$$= -\frac{1}{5}$$

15. Trigonometry, EXT1 T3 SM-Bank 12

$$2 \sin x \cos x = \sin x$$

$$\sin x(2 \cos x - 1) = 0$$

$$\sin x = 0, \cos x = \frac{1}{2}$$

$$\text{If } \sin x = 0 \Rightarrow x = 0, \pi, 2\pi$$

$$\text{If } \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

MARKER'S COMMENT: Many students expanded $\sin(2x)$ and then cancelled $\sin x$ on both sides which lost a set of solutions!

16. Trigonometry, EXT1 T3 SM-Bank 8

$$x \sin(x) \sec(2x) = \frac{x \sin(x)}{\cos(2x)} = 0$$

Find x that satisfies:

$$x = 0 \text{ and } \cos(2x) \neq 0 \Rightarrow x = 0$$

or,

$$\sin(x) = 0 \text{ and } \cos(2x) \neq 0 \Rightarrow x = \pi, 2\pi$$

$$\therefore x = 0, \pi \text{ or } 2\pi$$

17. Trigonometry, EXT1 T3 2020 HSC 11d

$$A \sin(x + \alpha) = \sqrt{3} \sin x + 3 \cos x$$

$$A \sin x \cos \alpha + A \cos x \sin \alpha = \sqrt{3} \sin x + 3 \cos x$$

Equating co-efficients:

$$\Rightarrow A \cos \alpha = \sqrt{3}$$

$$\Rightarrow A \sin \alpha = 3$$

$$A^2 = (\sqrt{3})^2 + 3^2 = 12$$

$$\therefore A = \sqrt{12}$$

$$\frac{A \sin \alpha}{A \cos \alpha} = \frac{3}{\sqrt{3}}$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$\sqrt{3} \sin x + 3 \cos x = \sqrt{3}$$

$$\sqrt{12} \sin\left(x + \frac{\pi}{3}\right) = \sqrt{3}$$

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{3} = \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$\therefore x = \frac{\pi}{2}, \frac{11\pi}{6} \quad (0 \leq x \leq 2\pi)$$

18. Trigonometry, EXT1 T3 2021 HSC 11g

$$2\sin^3 x + 2\sin^2 x - \sin x - 1 = 0$$

$$2\sin^2 x(\sin x + 1) - (\sin x + 1) = 0$$

$$(2\sin^2 x - 1)(\sin x + 1) = 0$$

$$2\sin^2 x = 1 \quad \sin x = -1$$

$$\sin^2 x = \frac{1}{2}$$

$$\sin x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

19. Trigonometry, EXT1 T3 EQ-Bank 3

$$\sqrt{3}\cos\theta = \sin(2\theta)$$

$$2\cos\theta\sin\theta - \sqrt{3}\cos\theta = 0$$

$$\cos\theta\left(\sin\theta - \frac{\sqrt{3}}{2}\right) = 0$$

$$\cos\theta = 0 \text{ or } \sin\theta = \frac{\sqrt{3}}{2}$$

COMMENT: Expressing 'all values' is specifically mentioned in Topic Guidance as an application of arithmetic sequences.

$$\theta = 90^\circ, 270^\circ \text{ or } \theta = 60^\circ, 120^\circ \quad (0^\circ \leq \theta \leq 360^\circ)$$

$$\therefore \theta = 90^\circ + m \times 180^\circ, 60^\circ + m \times 360^\circ \text{ or } 120^\circ + m \times 360^\circ$$

where m is an arbitrary integer.

20. Trigonometry, EXT1 T3 EQ-Bank 4

i. $f(t) = 6\sin 0.05t + 8\cos 0.05t$

$$A\sin(at + b) = A\sin(0.05t + b)$$

$$= A\sin(0.05t)\cos(b) + A\cos(0.05t)\sin(b)$$

$$\Rightarrow A\cos(b) = 6, A\sin(b) = 8$$

$$A^2 = 6^2 + 8^2$$

$$A = 10$$

$$\Rightarrow 10\cos(b) = 6$$

$$\cos(b) = \frac{6}{10}$$

$$b = \cos^{-1} 0.6 \approx 0.927 \text{ radians}$$

$$\therefore f(t) = 10\sin(0.05t + 0.927)$$

ii. Max occurs at $\sin(0.05t + 0.927) = \sin \frac{\pi}{2}$

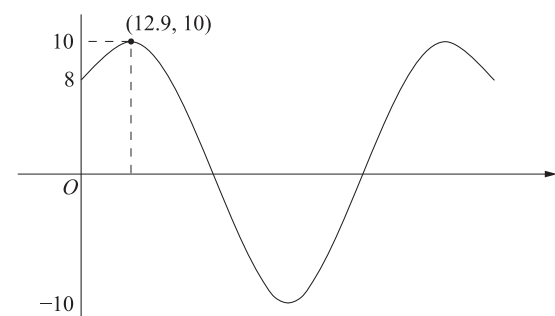
$$0.05t + 0.927 = \frac{\pi}{2}$$

$$0.05t = 0.643\dots$$

$$\therefore t = 12.87$$

$$= 12.9 \text{ (to 1 d.p.)}$$

iii.



COMMENT: High achieving students know the 3 variants of **$\cos 2x$** back to front. Here, **$\cos 2x = \cos^2 x - \sin^2 x$** breaks the back of this problem.

$$\text{LHS} = \cos(2x + x)$$

$$= \cos 2x \cos x - \sin 2x \sin x$$

$$= (\cos^2 x - \sin^2 x) \cos x - 2 \sin x \cos x \sin x$$

$$= \cos^3 x - \sin^2 x \cos x - 2 \sin^2 x \cos x$$

$$= \cos^3 x - (1 - \cos^2 x) \cos x - 2(1 - \cos^2 x) \cos x$$

$$= \cos^3 x - \cos x + \cos^3 x - 2 \cos x + 2 \cos^3 x$$

$$= 4 \cos^3 x - 3 \cos x$$

22. Trigonometry, EXT1 T3 2010 HSC 4b

i. $2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right)$

$$= 2 \cos \theta + 2 \left(\cos \theta \cos \left(\frac{\pi}{3} \right) - \sin \theta \sin \left(\frac{\pi}{3} \right) \right)$$

$$= 2 \cos \theta + 2 \cos \theta \times \frac{1}{2} - 2 \sin \theta \times \frac{\sqrt{3}}{2}$$

$$= 2 \cos \theta + \cos \theta - \sqrt{3} \sin \theta$$

$$= 3 \cos \theta - \sqrt{3} \sin \theta$$

$$R \cos(\theta + \alpha) = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$R \cos \alpha = 3 \quad R \sin \alpha = \sqrt{3}$$

$$\cos \alpha = \frac{3}{R} \quad \sin \alpha = \frac{\sqrt{3}}{R}$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6} \quad \left(0 < \alpha < \frac{\pi}{2} \right)$$

$$R^2 = 3^2 + (\sqrt{3})^2$$

$$= 9 + 3$$

$$R = \sqrt{12} = 2\sqrt{3}$$

$$\therefore 2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 2\sqrt{3} \cos \left(\theta + \frac{\pi}{6} \right)$$

ii. $2 \cos \theta + 2 \cos \left(\theta + \frac{\pi}{3} \right) = 3$

$$2\sqrt{3} \cos \left(\theta + \frac{\pi}{6} \right) = 3$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\cos^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{6}$$

♦ Mean mark part (ii) 49%
MARKER'S COMMENT: Many students did not check their answers against the stated domain for θ .

Since $\cos$ is positive in 1st and 4th quadrants,

$$\theta + \frac{\pi}{6} = \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$\therefore \theta = \frac{5\pi}{3} \quad (0 < \theta < 2\pi)$$

23. Trigonometry, EXT1 T3 EQ-Bank 1

i. $\sin x + \sin 3x = \sin x + \sin 2x \cos x + \cos 2x \sin x$

$$= \sin x + 2 \sin x \cos^2 x + (\cos^2 x - \sin^2 x) \sin x$$

$$= \sin x + 2 \sin x \cos^2 x + \cos^2 x \sin x - \sin^3 x$$

$$= \sin x + 3 \sin x (1 - \sin^2 x) - \sin^3 x$$

$$= \sin x + 3 \sin x - 3 \sin^3 x - \sin^3 x$$

$$= 4 \sin x (1 - \sin^2 x)$$

$$= 4 \sin x \cos^2 x$$

$$= 2 \sin 2x \cos x$$

$$= \text{RHS}$$

ii. $\sin x + \sin 2x + \sin 3x = 0$

$$\sin 2x + 2 \sin 2x \cos x = 0$$

$$\sin 2x (1 + 2 \cos x) = 0$$

If $\sin 2x = 0$:

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\therefore x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

If $\cos x = -\frac{1}{2}$:

$$\therefore x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

24. Trigonometry, EXT1 T3 EQ-Bank 2

i. $f(t) = \sqrt{5} \sin 0.2t + 2 \cos 0.2t$

$$\begin{aligned} R \sin(nt - \alpha) &= R \sin(0.2t - \alpha) \\ &= R \sin 0.2t \cos \alpha - R \cos 0.2t \sin \alpha \end{aligned}$$

$$\Rightarrow R \cos \alpha = \sqrt{5}, \quad R \sin \alpha = -2$$

$$R^2 = (\sqrt{5})^2 + (-2)^2 = 9$$

$$R = 3$$

$$\cos \alpha = \frac{\sqrt{5}}{3}, \quad \sin \alpha = -\frac{2}{3}$$

$$\Rightarrow \alpha \text{ is in 4th quadrant}$$

$$\text{Base angle} = \cos^{-1}\left(\frac{\sqrt{5}}{3}\right) = 0.7297$$

$$\begin{aligned} \therefore \alpha &= 2\pi - 0.7297 \\ &= 5.553... \end{aligned}$$

$$\therefore f(t) = 3 \sin(0.2t - 5.553)$$

ii. Max value occurs when $\sin(0.2t - 5.553) = 1$

$$0.2t - 5.553 = \frac{\pi}{2}$$

$$0.2t = 7.124...$$

$$t = 35.62...$$

Test if $t > 0$ for:

$$0.2t - 5.553 = -\frac{3\pi}{2}$$

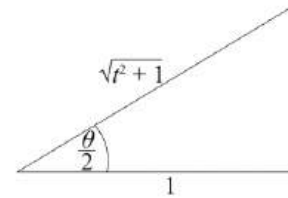
$$0.2t = 0.8406$$

$$t = 4.20...$$

$$\therefore \text{Time of 1st maximum: } t = 4.2$$

25. Trigonometry, EXT1 T3 2005 HSC 4b

Show $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$



$$\tan \frac{\theta}{2} = t$$

$$\Rightarrow \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = \frac{2t}{t^2 + 1}$$

$$\Rightarrow \cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \frac{1 - t^2}{t^2 + 1}$$

$$\Rightarrow \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{2t}{1 - t^2}$$

$$\begin{aligned} \text{LHS} &= \frac{1}{\sin \theta} + \frac{1}{\tan \theta} \\ &= \frac{t^2 + 1}{2t} + \frac{1 - t^2}{2t} \\ &= \frac{t^2 + 1 + 1 - t^2}{2t} \\ &= \frac{1}{t} \\ &= \frac{1}{\tan \frac{\theta}{2}} \\ &= \cot \frac{\theta}{2} \\ &= \text{RHS} \dots \text{ as required.} \end{aligned}$$

26. Trigonometry, EXT1 T3 2008 HSC 6b

Substitute $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$
into $\sin 3\theta + \sin 2\theta = \sin\theta$,

$$3\sin\theta - 4\sin^3\theta + \sin 2\theta = \sin\theta$$

$$2\sin\theta - 4\sin^3\theta + 2\sin\theta\cos\theta = 0$$

$$2\sin\theta[1 - 2\sin^2\theta + \cos\theta] = 0$$

$$2\sin\theta[1 - 2(1 - \cos^2\theta) + \cos\theta] = 0$$

$$2\sin\theta[1 - 2 + 2\cos^2\theta + \cos\theta] = 0$$

$$2\sin\theta[2\cos^2\theta + \cos\theta - 1] = 0$$

$$2\sin\theta(2\cos\theta - 1)(\cos\theta + 1) = 0$$

$$2\sin\theta = 0$$

$$\Rightarrow \theta = 0, \pi, 2\pi$$

$$2\cos\theta - 1 = 0$$

$$\cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

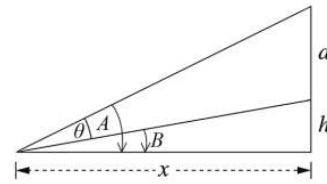
$$\cos\theta + 1 = 0$$

$$\cos\theta = -1$$

$$\Rightarrow \theta = \pi$$

$$\therefore \theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi \quad (0 \leq \theta \leq 2\pi)$$

27. Trigonometry, EXT1 T3 SM-Bank 1



$$\text{Show } \theta = \tan^{-1} \left[\frac{ax}{x^2 + h(a+h)} \right]$$

$$\tan A = \frac{a+h}{x}$$

$$\tan B = \frac{h}{x}$$

$$\begin{aligned} \tan(A - B) &= \frac{\frac{a+h}{x} - \frac{h}{x}}{1 + \left(\frac{a+h}{x}\right)\left(\frac{h}{x}\right)} \times \frac{x^2}{x^2} \\ &= \frac{x(a+h) - xh}{x^2 + h(a+h)} \\ &= \frac{ax}{x^2 + h(a+h)} \end{aligned}$$

$$\text{Since } \theta = A - B$$

$$\theta = \tan^{-1} \left[\frac{ax}{x^2 + h(a+h)} \right] \quad \dots \text{ as required.}$$

MARKER'S COMMENT: Answers that included a diagram and clearly labelled angles were generally successful.

i. Prove: $\sin^3 \theta - \frac{3}{4}\sin \theta + \frac{\sin(3\theta)}{4} = 0$

$$\begin{aligned}\text{LHS} &= \sin^3 \theta - \frac{3}{4}\sin \theta + \frac{1}{4}(\sin 2\theta \cos \theta + \cos 2\theta \sin \theta) \\ &= \sin^3 \theta - \frac{3}{4}\sin \theta + \frac{1}{4}(2\sin \theta \cos^2 \theta + \sin \theta(1 - 2\sin^2 \theta)) \\ &= \sin^3 \theta - \frac{3}{4}\sin \theta + \frac{1}{4}(2\sin \theta(1 - \sin^2 \theta) + \sin \theta - 2\sin^3 \theta) \\ &= \sin^3 \theta - \frac{3}{4}\sin \theta + \frac{1}{4}(2\sin \theta - 2\sin^3 \theta + \sin \theta - 2\sin^3 \theta) \\ &= \sin^3 \theta - \frac{3}{4}\sin \theta + \frac{3}{4}\sin \theta - \sin^3 \theta \\ &= 0\end{aligned}$$

ii. Show $\sin(3\theta) = \frac{1}{2}$

Using part (i):

$$\begin{aligned}\frac{\sin(3\theta)}{4} &= \frac{3}{4}\sin \theta - \sin^3 \theta \\ \sin(3\theta) &= 3\sin \theta - 4\sin^3 \theta \dots (1)\end{aligned}$$

$$x^3 - 12x + 8 = 0$$

Let $x = 4\sin \theta$

$$(4\sin \theta)^3 - 12(4\sin \theta) + 8 = 0$$

$$64\sin^3 \theta - 48\sin \theta = 0$$

$$\underbrace{-16(3\sin \theta - 4\sin^3 \theta)}_{\text{see (1) above}} = -8$$

$$-16\sin(3\theta) = -8$$

$$\sin(3\theta) = \frac{1}{2}$$

iii. Prove: $\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18} = \frac{3}{2}$

Solutions to $x^3 - 12x + 8 = 0$ are

$$x = 4\sin \theta \text{ where } \sin(3\theta) = \frac{1}{2}$$

When $\sin 3\theta = \frac{1}{2}$,

$$3\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}, \dots$$

$$\theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}, \dots$$

$\therefore$ Solutions

$$x = 4\sin \frac{\pi}{18} \quad \left(= 4\sin \frac{17\pi}{18} \right)$$

$$x = 4\sin \frac{5\pi}{18} \quad \left(= 4\sin \frac{13\pi}{18} \right)$$

$$x = 4\sin \frac{25\pi}{18} \quad \left(= 4\sin \frac{29\pi}{18} \right)$$

If roots of $x^3 - 12x + 8 = 0$ are α, β, γ :

$$\alpha + \beta + \gamma = -\frac{b}{a} = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = -12$$

$$\left(4\sin \frac{\pi}{18}\right)^2 + \left(4\sin \frac{5\pi}{18}\right)^2 + \left(4\sin \frac{25\pi}{18}\right)^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$16\left(\sin^2 \frac{\pi}{18} + \sin^2 \frac{5\pi}{18} + \sin^2 \frac{25\pi}{18}\right) = 0 - 2(-12)$$

$$\therefore \frac{\sin^2 \pi}{18} + \sin^2 \frac{5\pi}{18} + \frac{\sin^2(25\pi)}{18} = \frac{24}{16} = \frac{3}{2}$$